DYLAN CHEN

dylanchen4306@gmail.com | 929-488-9693 | LinkedIn | Portfolio

EDUCATION

Northeastern University - *Boston, MA* *Expected May 2028*
Bachelor of Science in Computer Engineering and Computer Science *GPA: 3.5/4.0*
Relevant Coursework: Algorithms and Data, Intro to Databases, Embedded Systems, Digital Design & Computer Organization, Networks, Discrete Math, Computing Fundamentals
Activities & Leadership: UX Designer *(Sandbox NU)*, Software Developer *(Oasis Accelerator)*

SKILLS

- **Languages & Embedded:** TypeScript/JavaScript, Python, C/C++, Java, Kotlin, SQL, HTML/CSS, R, ESP32, Arduino, UART
- **Front End & Full Stack:** React, Astro, Three.js, HTML5 Canvas, REST APIs, MySQL, SQLite, Aiven (Cloud DB), R Shiny
- **HCI & Design:** Figma, Wireframing, Prototyping, Usability Iteration, Jetpack Compose (Android), Blender
- **Tools & AI:** Git/GitHub, GitHub Actions (CI/CD), Jira, Agile/Scrum, Claude Code, Claude MCP (Figma, CAD)

TECHNICAL EXPERIENCE

Embedded Engineer Co-Op (Incoming) - *Bevi* *Starting January 2027*

Product Test Engineer Co-op - *SharkNinja* *January 2026 - June 2026*

- Programmed Arduino (C++) firmware with interactive, menu-driven operator UIs for custom test rigs, automating manual life-cycle testing and raising daily test throughput by 600%.
- Accelerated GUI and dashboard iteration through rapid prototyping, using direct operator feedback to set UX goals and deliver an MVP for test protocol deployment.
- Built a Python image-analysis tool using Otsu thresholding that auto-generates PowerPoint and Excel deliverables, cutting post-processing analysis time by 98%.
- Authored technical documentation and led training for directors, managers, and operators on custom rigs and tools.

Remote Support Engineer Intern - *Insights Gaming* *August 2023 - October 2023*

- Diagnosed and resolved 50+ support tickets by reproducing bugs and isolating root causes, tracking issues in Jira and writing 10+ Markdown troubleshooting guides.

PROJECTS

Android Drawing App *August 2026 - Present*

- Building an open-source native Android drawing app in Kotlin and Jetpack Compose with Claude Code, featuring tunable professional tool layouts and MyPaint brush engine integration.
- Tested and iterated UI/UX interactions and visuals to optimize user flow, informed by case-study research of existing creative software.

4-Degree-of-Freedom (4DOF) Robotic Arm *July 2026 - Present*

- Developing a web control hub in JavaScript with Three.js 3D rendering for an ESP32-based wireless robotic arm, structured for upcoming inverse kinematics and PID control.

Voice-Controlled Web Game (WALK-LOUDER) *May 2026 - Present*

- Built a TypeScript/Phaser platformer that maps live microphone loudness to player movement using FFT analysis, noise-floor calibration, and low-pass smoothing.

Smart Control Knob (NOB) *September 2025 - Present*

- Built ESP32-C3 C++ firmware driving a round SPI display with live volume feedback, plus a Python host app over UART that controls Windows volume in real time.

Cheese Affinage Database - *Northeastern University* *August 2026*

- Designed a 3NF MySQL database from an ERD and deployed it on Aiven (cloud-hosted), with an accessible R Shiny dashboard for querying and visualizing data.

Personal Portfolio Website *May 2025 - July 2026*

- Architected a portfolio in Astro and TypeScript with Claude Code, adding SPA-like view transitions, Canvas and SVG physics-based animations, drag-and-drop interactions, and serverless EmailJS integration.

INTERESTS

UI/UX, photographing wallpapers, coffee brewing, baking without measurements, climbing rocks, doodles, and noodles